Care Creations™



Beauty Creations

Scientific Brochure

PatcH$_2$O®

The hydra-protect technology

BASF
We create chemistry

PatcH$_2$O®

The hydra-protect technology

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SUMMARY FILE

$PatcH_2O$®	A000297
Origin - Description	$PatcH_2O$® is a molecular patch based on film-forming natural biopolymers (hyaluronic acid, alginate and pullulan) carrying highly concentrated hydrating substances (trehalose, glycerin, urea and L-serine)
Regulatory data	
INCI	Water (and) Glycerin (and) Trehalose (and) Urea (and) Serine (and) Pentylene Glycol (and) Glyceryl Polyacrylate (and), Algin (and), Caprylyl Glycol (and) Sodium Hyaluronate (and), Pullulan (and) Disodium Phosphate (and) Potassium Phosphate
China	Each component is listed in Inventory of Existing Cosmetic Ingredient in China (2015) IECIC
CAS#	7732-18-5, 56-81-5, 99-20-7, 57-13-6, 56-45-1, 5343-92-0, 104365-75-5, 9005-38-3, 1117-86-8, 9067-32-7, 9057-02-7, 7558-79-4, 7778-77-0
EINECS#	231-791-2, 200-289-5, 202-739-6, 200-315-5, 200-274-3, 226-285-3, 232-680-1, 214-254-7, 232-678-0, 232-945-1, 231-448-7, 231-913-4
Appearance	Colorless to yellow, clear to opalescent liquid, with characteristic odor
Preservative	None
Natural labels	None
Cosmetic uses	
Properties	• Powerful 'anti-thirst' hydration, for immediate and lasting effect • Rapid water boost for thirsty skin • Restores the quality of the cutaneous barrier • Inhibits cytoxicity induced by particulate matter (PM2.5) • Inhibits oxidative stress induced by Polycyclic Aromatic Hydrocarbon (PAH), e.g BaP • Decreases pollution remanence on skin. • Protects beneficial skin microbiota from toxic effect of pollution.
Applications	• Hydrating body milk • Express refreshing serum • Lasting comfort hydrating treatment • Soothing dry skin prevention lotion • Hydrating shampoo and gel • Cleanser for polluted skin
Formulation data	
Concentration of use	1 to 3%
Solubility	Soluble in water
Incorporation method	Incorporate into the aqueous phase during the formulation process up-to 85°
Optimal pH	4 - 8
Storage	10-30°C, protect against heat and from the effects of light
Shelf life	12 months
Patent family	Application filed

® = trademark of BASF in several countries



THE KEYS TO HYDRATION AND AWARENESS OF POLLUTION

Consumers cannot be wrong

In 2014/2015, hydration still represents the second largest cosmetics skin care market in terms of sales behind anti-ageing.

Market survey (Mintel) has identified growing consumer awareness of pollution to be a key trend for 2015.

Skin hydration mainly works at the surface but has a very strong impact on all of the properties of the skin: radiance, elasticity, youth and comfort are the real issues of hydration. Thus, maintaining and providing skin hydration when it's needed is of outmost importance.

Superficial but not superfluous

Talking about skin hydration is above of all else talking about hydration of the *stratum corneum* (the cornea layer) and the superficial layers of the epidermis. Under physiological conditions, this is normally ensured by the NMF[1], Natural Moisturizing Factor.
A complex blend of amino acids, minerals and osmolytes, NMF has a phenomenal water retention capacity. On its own, NMF ensures half of the water retaining capacities of the *stratum corneum*[2]. Moreover, it acts on the surface, because most of its components are found no more than 60 µm from the surface[3].
NMF is obviously not the only natural hydration factor. Hyaluronic acid (HA) also has a role to play through its hygroscopic power[4]. There still remains free water, or associated with other proteins with a higher molecular weight, like keratin.

Pollution affects skin beauty, quality and health. Pollution is made of different components like organic molecules, particulate matters and gasses. Pollution damages skin and induces dryness, premature ageing, dull skin, clogged pores, skin irritation, inflammation, and allergies[5,6].

Challenges and strategies

Once we know the importance and effectiveness of the NMF and the HA in skin hydration, what could be more obvious than to apply them directly onto the skin?
Cosmetics have thus used a cocktail of ingredients mimicking that of the NMF or the HA for decades in order to bring to the outside what the skin produces normally inside with obvious success.
Topical application of NMF components reduces transepidermal water loss (TEWL) and increases the capacity to retain moisture in the skin, and this within half an hour following application[7].
But what can we expect more than well moisturized skin? By improving skin hydration, we improve not only the barrier effect and the natural balance of the skin but also numerous parameters related to its appearance and youthfulness.

Well moisturized skin is first of all softer, more comfortable, a skin that "doesn't pull". It is also a skin that is supple, elastic and far less wrinkled[9,10,11,12]. Generally, well moisturized skin is skin that looks younger. It has even been shown that women who "do not look their age" are those that have the best hydration[13]. Finally moisturized skin is also a skin with improved sensory perception and finer microrelief[14]. And finally, a well moisturized skin better fulfils its important role as a barrier, protecting our body against environmental pollutants.

So we know all the benefits of moisturizing on one hand, and on the other we know effective moisturizers. So where are the new challenges for moisturizing products? The main challenge is clearly to achieve a balance between immediate and lasting hydration. Indeed, the effectiveness of the components of NMF applied topically is certainly quick but unsustainable.
Pollution is another challenge for which different strategies can be used such as:
- removing the pollutants by cleansing,
- applying antioxidants on the skin to protect from free radicals (ROS),
- adding foundation for an even skin tone,
- protecting against UV radiation.

Table 1 - Chemical composition of Natural Moisturizing Factor[8]

Chemical	Composition (%)
Free amino acids	40
Pyrrolidone carboxylic acid	12
Lactate	12
Sugars	8.5
Urea	7
Chloride	6
Sodium	5
Potassium	4
Ammonia, uric acid, glucosamine	1.5
Calcium	1.5
Magnesium	1.5
Phosphate	0.5
Citrate and formate	0.5

Overall, the best is to have an ingredient with anti-pollution capacity that will correct also some negative feature induced by pollution like dehydration.

So, for us, the challenge, is first to optimize the use of hydrating components by extending their bio-availability and slowing down their diffusion. In brief, control release. Second, providing skin hydration is necessary, but protecting skin cells and increasing skin barrier from external aggression such as pollution is even better.

A widely experienced expert in the vectorization and controlled release of active cosmetic ingredients, BASF Beauty Creations, with PatchH_2O® offers an efficient technical solution to meet the needs of two rapidly expanding market of hydration and anti-pollution ingredients.

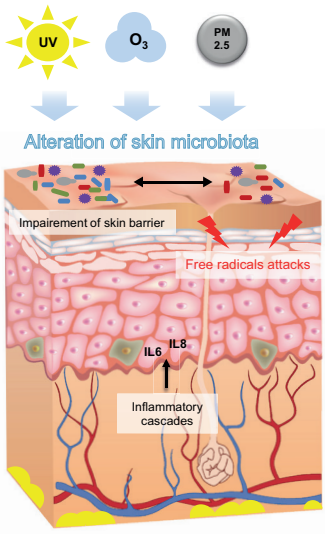


SKIN AND POLLUTION

Skin is the largest organ of the body and the critical protective barrier between our body and the environment. The hydration of skin is essential for its attractive appearance and helps to reinforce its barrier properties against deleterious effect of exogenous environmental factors[15]. Unfortunately, our skin is one of the first and major target of external assaults such as UV and air pollutants. Air pollutants include those of environmental origin, as well as those of anthropic origin[16]. Pollution is everywhere in cities, outdoor and indoor. Major air pollutants with effects on the skin include polycyclic aromatic hydrocarbons (PAHs), volatile organic compounds (VOCs), nitrogen oxides (NOx), particulate matter (PM), and cigarette smoke[17].

Air pollutants effect may be amplified in the presence of other air pollutants and in interaction with UV rays outdoor, and form major active components of the pro-oxidant smog. The protective ability of the skin is not unlimited, and problems arise when an abnormal exposure to environmental stressors exceeds the skin's defensive potential[18].

Air pollutants may induce severe interference of normal functions of lipids, DNA and/or proteins of the human skin via oxidative damage[17], leading to skin ageing. Ambiant air pollutant have been shown to alter skin microflora. Mancebo and coworkers showed that *in vivo* human skin exposure to ozone led to a 50% reduction in resident skin microflora suggesting that ozone has bactericidal effects[19].



Effects of pollution on skin.

Polycylic Aromatic Hydrocarbon (PAH) and Benzoapyrene (BaP)

The main source of atmospheric PAH, benzo[a]pyrene is residual wood burning. It is also found in automobile exhaust fumes and in all smoke resulting from the combustion of organic materials. PAH are highly lipophilic and can easily penetrate the skin[18]. They are converted into quinones, redox-cycling chemicals that produce reactive oxygen species[20,21] which in turn attack skin lipids, compromise skin barrier function and alter the skin capacity to retain water. In addition to that, BaP can impair keratinocytes proliferation and differentiation and finally epidermal homeostasis[22].

Particulate matters (PM)

PM are air pollutants, consisting of complex and varying mixtures of different size and composition particles suspended in the air. Factories, power plants, incinerators, automobiles, construction activities, fires and natural windblown dust are some of the main sources of PM[17] .Their major components are metals, organic compounds, material of biologic origin, ions, reactive gases, and the particle carbon core. Atmospheric PM with adsorbed metals (1648a) including particle size of 2.5µm (PM2.5) isolated from industrialized city has been shown to induce cytotoxicity toward skin cells and to decrease skin barrier function by increasing Transepidermal Water Loss (TEWL)[23]. The generation of oxidative stress by PM contributes to extrinsic skin aging[24]. Another study showed a significant correlation between PM pollution exposure (traffic particle and soot) and extrinsic skin aging signs, such as pigment spots on face, nasolabial folds, and wrinkles[24].

Generally, skin dryness increases when exposed to polluted environment[25].



THE HYDRA-PROTECT TECHNOLOGY

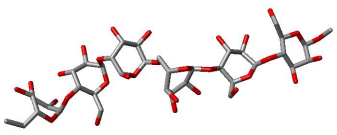
Conception

Thanks to natural bio-polymers such as pullulan, sodium alginate and hyaluronic acid, PatchH_2O® gives a new youthfulness to classic moisturizing ingredients like glycerin, urea, L-serine and trehalose. PatchH_2O® has a double function: it acts both as a protective film and as a hydration reserve.

Technology

The molecular mesh of PatchH_2O® is a unique combination of three polysaccharides - pullulan, alginate and hyaluronic acid - each bringing unique physiochemical properties.

Pullulan

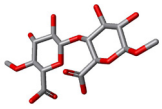


Pullulan is a linear polysaccharide composed of repeating maltotriose units (poly-α-1,6-maltotriose). This polymeric structure gives it unique gelling, adhesive and film forming properties[26]. Pullulan also has antioxidant benefits that, in combination with collagen, can increase the survival of stem cells placed in high oxidative stress and maintain them longer in their pluripotent state[27].

Pullulan is naturally produced by *Aureobasidium pullulans*, a microscopic ubiquitous non-pathogenic fungus able to adapt to many different environments thanks to its ability to withstand dehydration, irradiation and strong salinities. *Aureobasidium pullulans* is what we call an epiphyte: it grows on the surface of its host without causing damage, creating a barrier to protect against pathogens.

A. pullulans is thus an antagonist of pathogenic species such as *Aspergillus Niger* or *Botrytis cinerea*, which explains why it is used in the food industry to control the bacteriological environment and protect crops[28].

Alginate

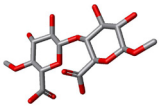


Alginic acid is a polymer of D-mannuronic and L-glucuronic units obtained by basic hydrolysis of brown algae (*Phaeophyceae*)[29].

Long used as pharmaceutical excipient for their gelling, stabilizing and film forming qualities, alginate gels have an ideal porosity that allows them to trap small molecules while retaining their free diffusion inside and outside of the gel.

Associated with certain drugs, alginate regulates the rate and kinetics of absorption of active ingredients.

Hyaluronic acid



Is there still need to describe it? Hygroscopic molecule par excellence - it can hold up to 1,000 times its volume in water - hyaluronic acid (HA) has emerged in recent years as the star of wrinkle filler injections.

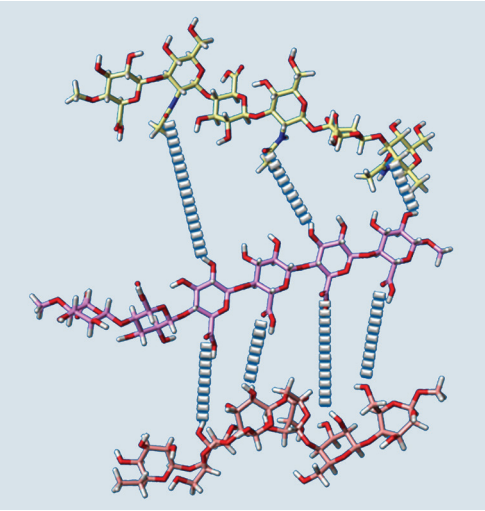


Figure 1 - 3D modeling of the "Hydrogen" interactions between Hyaluronic Acid, Alginate and Pullulan.

Despite its apparent simplicity of structure, this glycosaminoglycan stands out by the uniqueness of its functions: volume, migration, matrix assembly, healing[30,4], etc…
Without forgetting, of course, hydration. Half of the epidermal HA can be found in the *stratum corneum* where it interacts not only with water but also with phospholipids[31]. Thanks to its many free hydroxyl groups, hyaluronic acid is also capable of associating with other molecules to form complex networks. The injectable hyaluronic acids sold today are mostly cross-linked hyaluronic acid[1].

Combined in proportions of 1:2:1, pullulan, alginate and hyaluronic acid form a molecular mesh with the establishment of hydrogen bonds that enable the cohesion of the polymers up to dermal application.

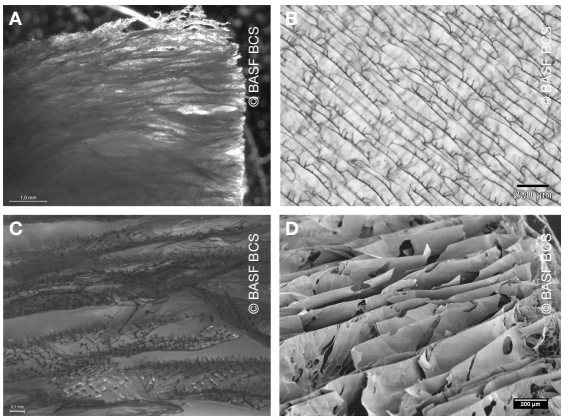
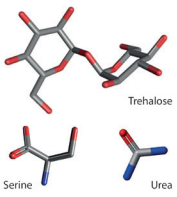


Figure 2 - Microscopic observations of PatchH2O®:
A and **B** - Macrofluorescence. **C** - Optical microscopy X10.
D - Transmission electron microscopy x60

This combination of polymers thus forms a film on the skin with a multi-layered structure that is easily observed after dehydration. Photos taken using the macro fluorescence technique show this very well organized multi-layer structure (Fig. 2A and 2B). At higher magnification, there are small bumps that are reminiscent of the micro-needles of new generation vectors (Fig. 2B). The network itself can be distinctly observed by optical microscopy (Fig. 2C). The layered structure is also clearly visible in scanning electron microscopy (Fig. 2D).

Hydrating complex



PatchH2O® would be nothing without the moisturizing complex that it contains: glycerin - serine - urea - trehalose. Serine and urea are both components of NMF. With its free hydroxyl group, serine binds easily with water molecules to form stable complexes[32]. Urea is itself a potent osmolyte actively involved in the regulation of the salinity of the extracellular environment[33].
Less known, trehalose is even more interesting. This disaccharide of glucose is mainly used today in the freezing solutions to improve the conservation of skin grafts[34,35]. During freezing, the water contained in the grafts forms crystals that alter the cell membranes and the overall quality of the tissue.

However, by binding to phospholipids, maintaining their spacing, forming a protective layer on the surface and sometimes even by taking the place of water molecules, trehalose stabilizes the membranes and finally preserves their integrity[36]. Finally, the addition of trehalose to cryo-preservation solutions to protect stem cells[37], artificial skin[38], skin grafts[39] and increases the chances of successful transplantation.
Another advantage of trehalose is in preventing denaturation by urea enzymes such as chymotrypsin, a serine protease similar to those found in the *stratum corneum* to ensure peeling.

Even if the composition is already unique in itself, the innovation lies above all in its performance: *in vitro, ex vivo, in vivo*, PatchH2O® balances a short, medium and long term moisturizing effect with rarely equaled effectiveness and additional anti-pollution efficacy.

THE HYDRA-PROTECT TECHNOLOGY, A HYDRIC BARRIER TO FIGHT POLLUTION

Safety / Tolerability of the product

In the recommended conditions of use, PatcH$_2$O® is not irritating the eyes or skin and there is no indication of skin sensitization and phototoxicity.

The moisturizing market consists of many products, but few of them today offer rapid and long-lasting hydration. On the dosage side, most of these products are composed of oily raw materials, leaving an uncomfortable feeling of stickiness on the skin.

Simple but effective. This is the motto that has guided the development of PatcH$_2$O®.

BASF Beauty Creations has chosen to combine its expertise in controlled release technology with its knowledge of biomacromolecules, in order to offer a product using a TECHNOLOGICAL INNOVATION that stands out from the competition.

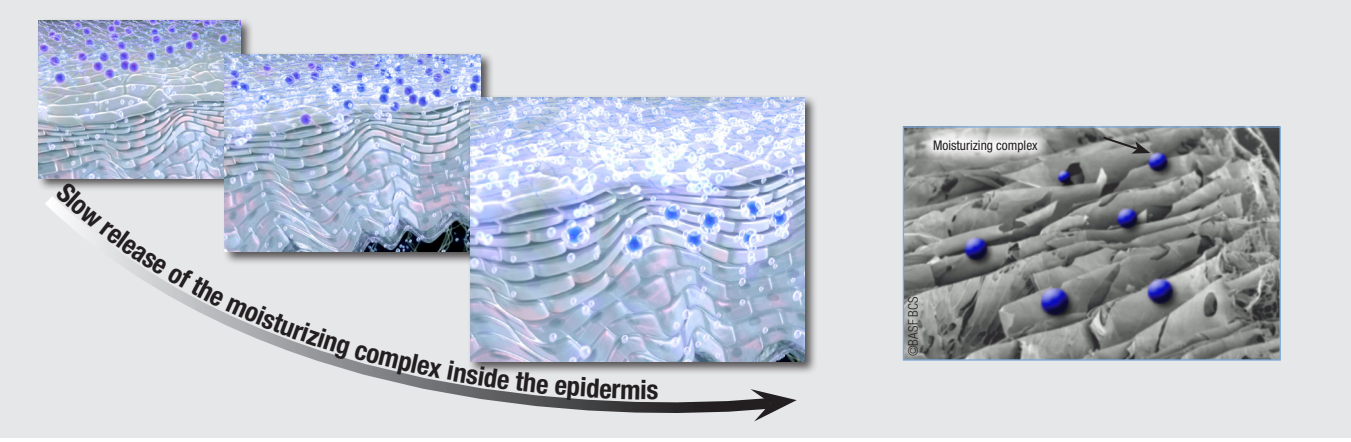
PatcH$_2$O® is a new moisturizing agent's concept, easily formulated, with a silky feel and an ability to really offer rapid and lasting hydration with concomitant protection against pollution.

PatcH$_2$O® developed by Beauty Creations perfectly merges all requirements of the moisturizing cosmetic market and the need for protection from increasing pollution.

Focus on technological innovation

Concept

PatcH$_2$O® is a molecular network of 3 natural biopolymers pullulan, alginate and hyaluronic acid transporting a highly concentrated hydration complex of glycerin, serine, trehalose and urea.



Immediate effect

The hydration complex brings its stabilizing power to the cell membranes to give the perfect finishing touch to the association and preserve the full integrity of the cells of the skin and the skin barrier. Moreover, PatcH$_2$O® limits pollution remanence on skin by enhancing skin cleansing after pollution deposition.

Long term effect

PatcH$_2$O® is designed to work on two levels for optimal hydration:
- on the surface of the skin, the alliance of its polymer film-forming powers forms a molecular mesh to preserve natural hydration of the skin,
- impregnated with moisturizing complex, this gradually releases into the heart of the superficial layers of the epidermis.

PatcH$_2$O® ensures sustained hydrations by decreasing transepidermal water loss while strengthening the skin's barrier function, which is daily attacked by pollution. Amazing physical properties of complex sugar, pullulan contributes both with its ability to form a film and its antioxidant properties[40], to fight oxidative stress induced by BaP combined with the film-forming properties of alginate and hyaluronic acid. The network thus formed can protect epidermal cell as well as skin beneficial microbiota from cytotoxicity induced by PM2.5 from industrialized city and acts as a reservoir that gradually distills its hydrating complex.

Easy to formulate

A colorless and fluid ingredient, PatcH$_2$O® has a composition specifically chosen for an optimum formulation. PatcH$_2$O® provides formulations with a silky smooth, non-greasy feel and leaves a sensation of comfort on the skin. PatcH$_2$O® has no contain preservatives.

Proven effectiveness

The efficiency tests prove it: *in vitro*, *ex vivo* or *in vivo*, the effectiveness of PatcH$_2$O® is indisputable.

Description

PatcH$_2$O® is an aqueous solution based on the innovative association of film-forming polymers and a powerful hydrating complex.



Example: Commercial sample of PatcH$_2$O® [1], incorporated respectively into a cream at 3% [2] or [3] hydrogel.



A DEMONSTRATED EFFICACY

By combining the film forming power of its polymers with the effectiveness of its hydrating complex, PatchH_2O® ensures immediate and lasting hydration. At the same time, this complex ensures protection of the skin against pollution.

Maintenance of water at the surface of the skin evaluated by

Aqua-guard power on synthetic skins during 48h.

Dielectric conductivity on human *stratum corneum* up to 48h.

Protection against pollution by limiting

Cytoxicity of PM2.5 on keratinocytes (ATP measurement).

Oxidative stress induced by BaP on keratinocytes (Malondialdehyde release).

Cytoxicity of PM2.5 on beneficial skin microbiota (*S. epidermidis*).

Remanence of charcoal on synthetic skin.

Gradual release of the moisturizing active complex into the surface layers of the epidermis through

Release of the urea and serine contained in PatchH_2O® in *stratum corneum* over 48h.

In vivo performance on

hydration test of the upper layers of the epidermis over 48h after a single application (corneometry).

Long term hydration test: 21 day up to 5 days after the last application (corneometry).

Self-evaluation questionnaire of panelists after twice daily treatment during 21 days.

EFFICACY

Aqua-guard power measurement

in vitro

OBJECTIVE

The purpose of this study was to show the ability of PatcH$_2$O® to maintain water at the level of the *stratum corneum* using an accelerated evaporation test on synthetic skins (Vitro-skin®).

RESULTS & DISCUSSION

The results showed that after 48 hours of conditions of evaporation at 45° C, the models of Vitro-skin® artificial skin treated with PatcH$_2$O® had retained significantly more water than skin treated with the only hydrating complex (+ 317%, p<0.001) (Figure 3).

CONCLUSION

Through its unique combination of film-forming polymers, PatcH$_2$O® promotes the maintenance of the water molecules in the skin for 48 hours (models of Vitro-skin®). These properties thus demonstrate the capacity of PatcH$_2$O® to offer prolonged hydration.

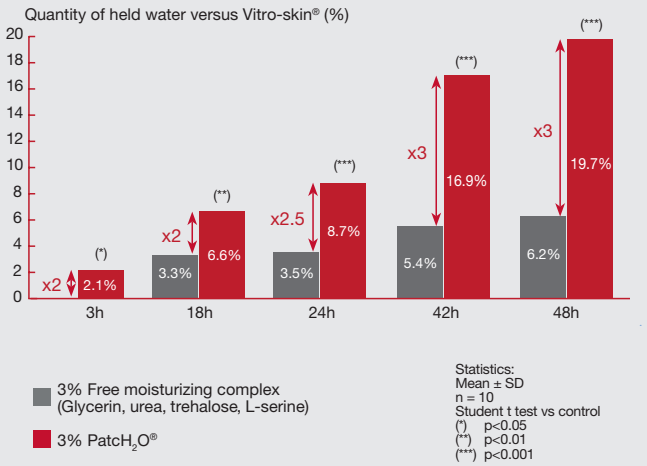


Figure 3 - Assessment of capacity of PatcH$_2$O® and hydrating complex only to limit the evaporation of water on synthetic skin (Vitro-skin®) over 48 hours.

MATERIALS&METHODS

Material

Vitro-skin® is a Mimetic artificial skin substrate used in many cosmetic tests. With the same topographies, pH, ionic strength and critical surface tension as human skin, it is a good model to study all active ingredients working on the properties of the skin's surface.

Treatment

PatcH$_2$O®, Moisturizing complex only (a mix of active moisturizer containing glycerin, urea, serine, and trehalose in water without polymers) and deionized (control) water are applied on 3 cm² of Vitro-skin® using a spreader rated at 200 µm thick. The treated skins are then dried at 45°C in a thermo-regulated oven for 1 hour in order to obtain a cohesive film on the surface.

Aqua-guard power measurement[41,42]

The processed and dried skins were positioned on containers containing approximately 6 cm^3 water, then placed again in the oven at 45° C for 48H of evaporation.

Aqua-guard power measurement was achieved by measuring the amount of water evaporated over the time. Each container's water loss was then measured by regular weighing. By simple difference, the Aqua-guard of PatcH$_2$O® was calculated in comparison with the moisturizing complex alone.

The results are expressed as a percentage of water maintained in comparison to the untreated control.

Statistics

The results come from 10 skins processed for each tested formula and expressed as Mean value ± Standard deviation.

Statistical analysis was performed using Student's t test, with a threshold of significance set at 5% (p<0.05).

EFFICACY / NEW

Effect of PatchH_2O® on oxidative stress induced by BaP combined with UVA

in vitro

OBJECTIVE

The aim of this study was to investigate the protective effect of PatchH_2O® against oxidative stress induced by Benzo[a] Pyrene (BaP) the most prevalent Polycyclic Aromatic Hydrocarbon (PAH) pollutant. In this study, BaP was combined with UVA to be closer to outdoor pollution conditions.

RESULTS & DISCUSSION

Malondialdehyde (MDA) is one of the final products of polyunsaturated fatty acids peroxidation in the cells. An increase in free radicals generated by BaP and its metabolites causes overproduction of MDA.

So, the oxidative stress has been measured by recording the levels of released MDA.

Vitamin E, an antioxidant widely used in cosmetics has been used as positive control (data not shown).

In the test conditions, UVA alone induced low oxidative stress. Applied together, BaP and UVA increased the MDA release significantly.

BaP and UVA together increase significantly MDA release. With PatchH_2O® starting at 0.062%, the level of MDA was significantly reduced (Figure 4 $p<0.01$ and $p<0.001$).

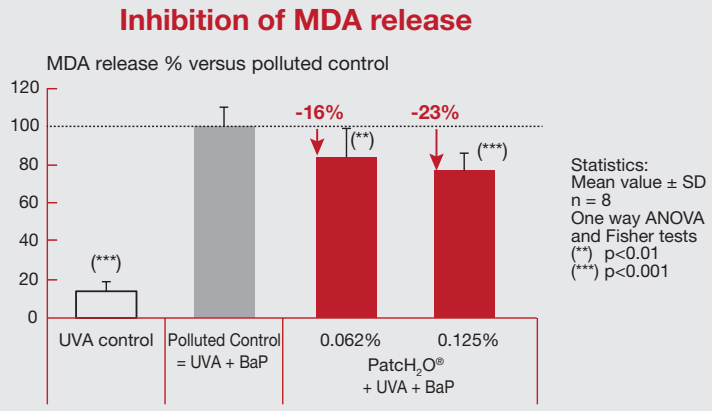


Figure 4 - Inhibition of MDA release by PatchH_2O®.

CONCLUSION

PatchH_2O® efficiently protects keratinocytes from deleterious effect of BaP pollutant. Starting at 0.062% PatchH_2O® protects skin from oxidative stress induced by BaP combined with UVA. This suggests that PatchH_2O® forms a barrier that limits penetration of BaP in cells and thus its prooxidant effect.

MATERIALS&METHODS

Cell culture

Human keratinocytes HaCaT cell line, were seeded within standard medium containing Fetal Calf serum (FCS) and incubated at 37°C, CO_2=5%, relative humidity >95% for 3 days in order to obtain a saturated cell layer.

Treatment

Cultured keratinocytes were treated by incubating them in medium with PatchH_2O® at 0.062% and 0.125% and Vitamin E (Sigma, France) as reference ingredient for one day. Then the cell culture medium was exchanged to a balanced salt solution (Hank's solution, Sigma France) containing BaP (1.5 µm) and then UVA irradiated (10 J/cm^2). Just after the irradiation, the quantity of released MDA was measured by the thiobarbituric acid (TBA, SIGMA, France) method. The amounts of protein were measured on cellular homogenate by the Bradford's method (Sigma, France), showing no direct cytotoxicity

Assay method: Thiobarbituric assay

0.5mL of each well containing sample, were transferred in glass tube, followed by 250µL of a 20% solution of Trichloroacetic acid (Sigma, France) and 250µL of a 2% solution of Thiobarbituric acid (Sigma, France). Samples were then incubated 30 minutes at 100°C, homogenized. 100µL of the resulting solution was mixed with 100µL of propanol-1 (Sigma, France), quantified by fluorescence at 532nm for excitation and 560nm for emission.

Results and statistics

Results are expressed as a percentage referring to UVA plus BaP irradiated condition and presented as a Mean ± SD (Standard Deviation) from 2 assays made in quadruplets (n=8). The statistics are based on ANOVA one way and "a posteriori" Fisher tests following Sigmaplot™ software recommendations.

EFFICACY / NEW

Effect of PatcH$_2$O® on cytotoxicity induced by particulate matter combined to UVA

in vitro

OBJECTIVE

The aim of this study was to evaluate the cytoprotective effect of PatcH$_2$O® exposed to PM2.5 (1648a). This pollution model is derived from the atmosphere of an industrialized city and contains PM2.5 as well as heavy metals. The cytoprotective effect has been assessed through the quantification of cellular viability via luminescent assay for ATP (Adenosine Triphosphate), which decreases when cells are damaged by cytotoxic component[43]. In this study, PM2.5 was combined with UVA to be closer to outdoor pollution conditions.

RESULTS & DISCUSSION

We have previously mentioned that UV potentiates the deleterious effect of pollutant. The polluted control made of cells incubated with PM2.5 (1648a) with UVA has significantly ($p<0.001$) lowered ATP production compaired to UVA alone, showing thus a cytotoxic effect of PM2.5. With PatcH$_2$O® at 0.25%, we succeeded in increasing significantly cellular ATP compared to the polluted control (Figure 5, $p<0.01$).

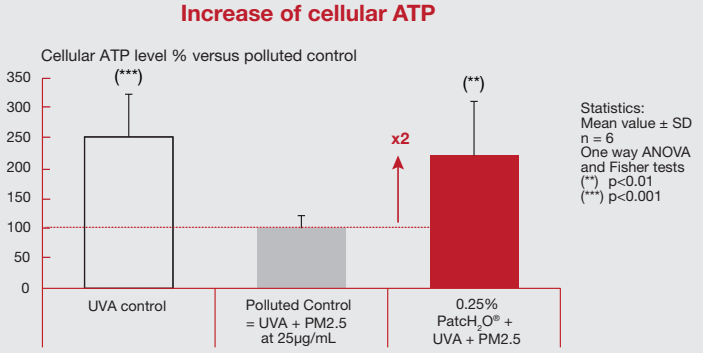


Figure 5 - Increase of cellular ATP with PatcH$_2$O® after incubation with PM2.5 and UVA.

CONCLUSION

PatcH$_2$O® at 0.25% can efficiently protect cells of our skin from damage, namely cytotoxicity induced by UVA and PM1648a which contains PM2.5 from industrialized city.

MATERIALS&METHODS

Material

Atmospheric particulate matter 1648a was obtained from NIST USA.

Atmospheric particulate matter 1648a was prepared from an industrialized city (St. Louis MO. USA) on a period of 12 months in 1976-1977 and contains metals such as aluminum 34 mg/g, iron 39 mg/g, lead 7 mg/g, zinc 4.8 mg/g, titanium 4 mg/g, chromium (0.4mg/g) arsenic (0.1mg/g). This particulate matter contains also organic molecules such as PAHs at lower concentration (50 µg/g), PCBs (0.18 µg/g) and Nitro-PAHs (0.3 µg/mg). Median diameter of particles is around 20µm but they contain a large quantity of PM < 10 µm and PM2.5.

Cell culture

Human keratinocytes HaCaT cell line, were seeded within standard medium containing Fetal Calf Serum (FCS; DUTSCHER France) and incubated at 37°C, and under 5% of CO_2, relative humidity >95% for 3 days in order to obtain a saturated cell layer.

Treatment

Cultured keratinocytes were treated by incubating them in balanced salt solution (Hank's solution, Sigma France) with particle matter 1648a (25 µm) and PatcH$_2$O® for 1 hour. The cell cultures were then irradiated by UVA (20 J/cm^2) and just after the irradiation, cell culture medium is renewed with standard medium DMEM (Invitrogen France) and incubated for 2 days at 37°C and under 5% CO_2.

Assay method: enzymatic measure of ATP

ATP Bioluminescence kit CLS 11699695001: ROCHE Diagnostics France

The luciferase from *Photinus pyralis* (American firefly) catalyzes the following reaction:

ATP + D-luciferin + O_2 + oxyluciferin + PPi + AMP + CO_2 + light.

The quantum yield for this reaction is about 90%. The resulting green light has an emission maximum at 562 nm.

The Michaelis equation has the following form: light intensity = (Vmax x CATP)/(Km + CATP).

At low ATP concentrations (CATP<Km), the formula is simplified to light intensity = Vmax x CATP/Km.

From this equation, it becomes obvious that the light output is directly proportional to the ATP concentration (CATP), and is dependent on the amount of luciferase (Vmax) present in the assay.

Samples and standard solution of ATP were transferred in multiwell plates followed by the same volume of Luciferase. The emitted light was then measured at 562nm and ATP concentration was deducted from the standard curve of ATP.

Results and statistics

Results are expressed as a percentage referring to polluted control (condition with 1648a PM plus UVA) and presented as a Mean ± SD (Standard Deviation). The statistics are based on ANOVA one way and "a posteriori" tests following Sigmaplot™ software recommendations.

EFFICACY / NEW

Effect of PatchH_2O® on skin cleansing after pollution particles deposition

in vitro

OBJECTIVE

The goal of this study was to evaluate the potential of PatchH_2O® to facilitate the washing of pollution particles from human skin.

RESULTS & DISCUSSION

During these experiments, synthetic skins ("Vitro-skin®") were used as skin models, and charcoal to mimic pollutants particles. Skin was pre-treated with PatchH_2O®, followed by a thin layer of charcoal particles. Then the amount of adherent charcoal particles was determined before and after washing with water by image analysis. PatchH_2O® at 1.5% in water has significantly improved the removal of charcoal particles from synthetic Vitro-skin® compared to the water control (Figure 6, +43% of particle removed with PatchH_2O®, p<0.01).

Considering the molecular network formed by PatchH_2O® polymers, we can assume that this improvement of cleansing is at least in part due to the trapping of charcoal particles in this network, which will be more easily removed by water washing

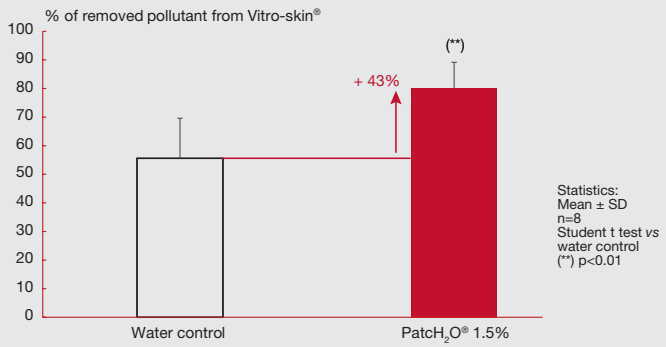


Figure 6 - PatchH_2O® cleansing effect on synthetic skin (Vitro-skin®) after charcoal deposition and washing.

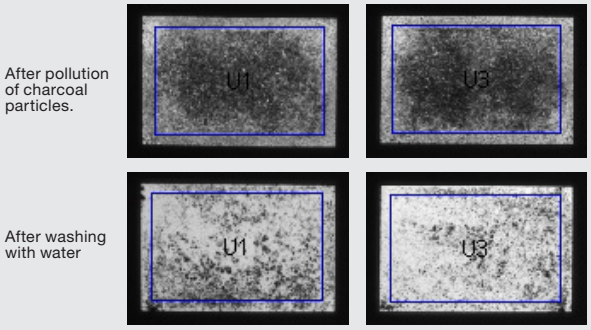


Figure 7 - Example of Vitro-skin® membrane after charcoal deposition and washing with water.

CONCLUSION

PatchH_2O® efficiently protects skin from deleterious effect of pollutant deposition and adherence by increasing their removal with water. PatchH_2O® decreases remanence of pollutant on skin by increasing its cleansing.

MATERIALS&METHODS

Treatment

Vitroskin™ (IMS Inc. Milford, CT, USA) membranes were cut in 4 X 4 cm squares and then hydrated by incubation for one day in a chamber containing a 15% glycerin (Sigma, France) and 85 % pure water solution. The hydrated squares were mounted in standard holder enabling a working surface of 3.5 cm X 2.5 cm (= 9 cm^2). PatchH_2O® at 1.5% in water was applied at 4 mg/cm^2 and incubated for 15 minutes in 40% humidity atmosphere. After that, 15 mg of charcoal (Pharmacy, France) were dusted on each treated square and the non-adherent particles are gently eliminated. Finally, each treated and non-treated-polluted square was washed by 3 slow passages in pure water and leave to dry for 1 hour in ambient atmosphere.

Assay method

The amount of charcoal particles on Vitro-skin® was assessed by image analysis using an automated gel imaging instrument (Chemidoc™ XRS Biorad, France).

A view of each treated square was recorded after charcoal deposition, and after washing with water.

The evaluation was based on grey levels determined automatically by gel imaging instrument for each square at every key steps, i.e. 1 = after treatment by product and water control (picture not shown), 2 = after pollution by charcoal particles, 3 = after washing in water. The level of removed charcoal is based on the difference of grey levels between 2 and 3, related to the difference of grey levels between 1 and 2.

Results and statistics

The assay is carried out in eight replicate and the results are expressed in percentage as a mean ± standard deviation (SD). Statistical analysis was performed using student t test, with a threshold of significance set at 5% (p<0.05).

EFFICACY / NEW

Effect of PatcH$_2$O® on skin microbiota after exposure to particulate matter

in vitro

OBJECTIVE

The aim of this study was to prove the toxicity of PM2.5 pollutants toward skin microbiota and the selective protective effect of PatcH$_2$O® to counteract this effect.

RESULTS & DISCUSSION

During this study, urban dust PM2.5 containing PAH were used as pollutant.

Resident cutaneous microbiota helps to maintain homeostasis and prevent overgrowth of harmful microbes. It is of importance to protect this skin resident microbiota[19]. *Staphylococcus epidermidis* has been used as an example of skin microbiota specimen as it is one of the most common microorganism found on the skin with proven benefits for the skin. While *Propionibacterium acnes* often involved in acne vulgaris is used as an opportunistic pathogenic microorganism.

We have shown that pollution and namely PM2.5 can affect skin microbiota balance by decreasing skin microbial species growth like *Staphylococcus hominis*, *Staphylococcus epidermidis* and *Propionibacterium* acnes (data shown only for *S. epidermidis*). This growth decrease is linked to cytotoxicity induced by PM2.5.

In this study Sodium Dodecyl Sulfate was successfully used as positive control of cytotoxicity to validate the experiment. Water showed an increase in bacterial growth after 24h and 72h respectively for *S. epidermidis* and *p. acnes*. PM2.5 in Dimethylsulfoxide (DMSO) showed a dramatic decrease in both bacteria growth. This growth was significantly rescued when *S. epidermidis* samples were treated at the same time with 0.5% of PatcH$_2$O® (Figure 8 x1.5 of higher growth with PatcH$_2$O®, compared to polluted control; p<0.01). With *P. acnes*, PatcH$_2$O® didn't show any significant protection against cytotoxicity induced by pollutant PM2.5 (Figure 9), protecting thus selectively beneficial skin microbiota.

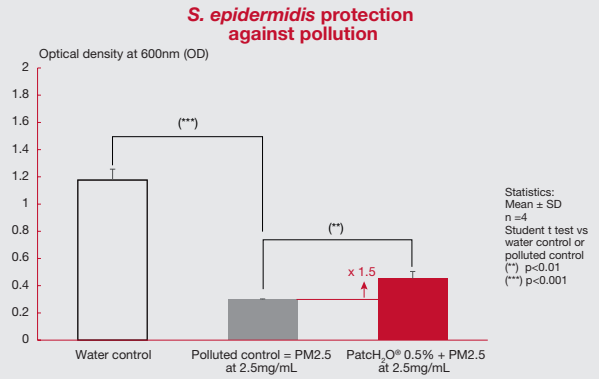


Figure 8 - *S. epidermidis* growth protection by PatcH$_2$O® against PM2.5 cytotoxicity.

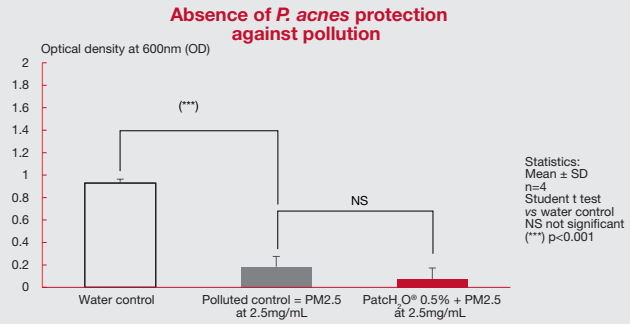


Figure 9 - Absence of *P. acnes* growth protection by PatcH$_2$O® against PM2.5 cytotoxicity.

CONCLUSION

PatcH$_2$O® can be used to preserve skin beneficial microbiota from deleterious effect of pollution, namely PM2.5. This helps to maintain skin beauty and homeostasis.

MATERIALS&METHODS

Bacteria culture

S. epidermidis ATCC 14990 was precultured 24h in a tube containing 5mL of TSB (Tryptic soy broth) under stirring in an incubator shaker (New Brunswick™, Innova 42®). While *P. acnes* ATCC6919 was cultured in anaerobic bag. Afterwards, 180µL of bacterial suspension were seeded at an Optical Density (OD) of 0.05 in TSB medium and at 0.3 in Shaedler medium respectively for *S. epidermidis* and *P. acnes* in 96 well plates. The volume per well is completed to 200µL, by adding 20µL of solution containing pollutant, product or a mixture of both. The resulting mixtures were incubated using an incubator microplate reader (Spark®10M, TECAN France).

Treatment

S. epidermidis suspension was incubated 24h under aerobic condition in presence of PM2.5 suspensions at 2.5mg/mL, or a mixture of PM2.5 at 2.5mg/mL and PatcH$_2$O® for treated conditions. For *P. acnes*, the incubation was made using the same conditions, but under anaerobic and with an extended period of 72 hours. DMSO and Water were used as solvent control respectively for pollutant PM2.5 (data not shown) and PatcH$_2$O®. Samples were tested in quadruplicate.

Assay method

After incubation period, bacterial growth was assessed by recording of OD at 600nm, at 37°C under shaking.

Results and statistics

The assay is carried out in quadruplicate and the results are expressed as a mean OD value ± standard deviation (SD). Statistical analysis was performed using student t test, with a threshold of significance set at 5% (p<0.05).

EFFICACY

Measurement of dielectric conductivity - Tagami Test

ex vivo

OBJECTIVE

The purpose of this study was to demonstrate the hydrating power of the complex of ingredients and the contribution of the network of polymers of PatchH_2O® in maintaining water at the level of the epidermis.
To do this, dielectric conductivity of a human skin biopsies treated with 3% of PatchH_2O® is compared with a biopsy treated with 3% hydrating complex only.

RESULTS & DISCUSSION

The moisturizing activity of PatchH_2O® formulated at 3% in a hydrogel was assessed by measurement of the dielectric conductivity. Inherently weakly hydrated, the *stratum corneum* is a substrate for low electrical conductivity, not very sensitive to the electric field due to the bipolarity of water molecules.
To sum up, more the *stratum corneum* is moisturized the higher the conductivity is.
The application of a hydrogel on its own naturally increases dielectric conductivity and therefore the state of hydration of the *stratum corneum*.
The results of PatchH_2O® and the hydrating complex alone was therefore expressed in percentage of increase in dielectric conductivity compared to hydrogel alone (control).
The results showed that PatchH_2O® formulated at 3% in a hydrogel increases the dielectric conductivity in the *stratum corneum* quite significantly compared to the placebo and this from 30 min up to 48 hours after application.
48 hours after treatment, the moisturizing power of PatchH_2O® is on average 1.6 times that of the moisturizing complex alone (Figure 10).

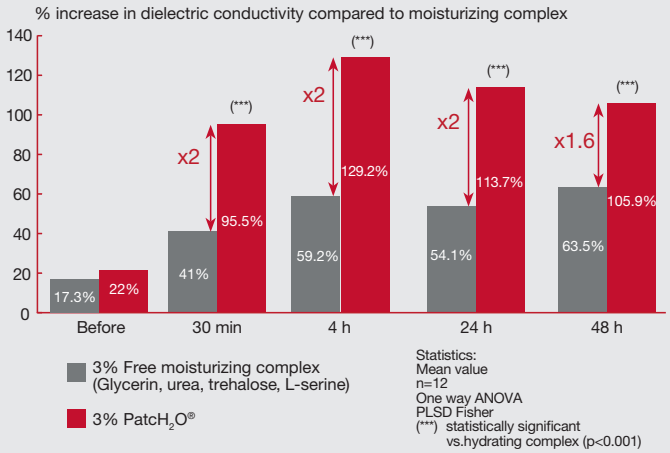


Figure 10 - The moisturizing power of PatchH_2O® and the moisturizing complex alone is expressed in percentage of increase in dielectric conductivity compared to hydrogel alone.

CONCLUSION

In this study, the moisturizing properties of PatchH_2O® have been demonstrated on human skin explants. PatchH_2O® offers intense moisturizing to the upper layers of the epidermis from 30 minutes and lasting up to 48 hours, and this efficiency is better than that offered by the placebo. The results of this study confirm the results of the previous study which demonstrates again the benefit of the polymer network of PatchH_2O®.

MATERIALS&METHODS

Material

We used Hygrometer model Skicon 200 (IBS Inc., Hamamatsu, Japan), which reads the conductance in terms of µS (microsiemen).

Treatment

This study was conducted with human skin explants from plastic surgery. Epidermal sheets were prepared by peeling of the dermis after heating at 60° C for 2 minutes. The *stratum corneum* was isolated by enzymatic digestion as described in the publication of Rochefort et al[43].

The epidermal sheets were balanced in an oven set at a relative humidity (44% - saturated solution of potassium carbonate), then were treated with Control Hydrogel (no active ingredients), Hydrogel containing the moisturizing complex only at 3% and Hydrogel containing PatchH_2O® at 3%. There was three consecutive applications spaced 30 minutes apart, applied doses were 1 mg/cm² (12 trials).

Assay method: Dielectric conductivity measurement[45,46]

Dielectric conductivity measurements were carried out prior to treatment and then up to 48 hours after the third application.
The moisturizing effect expressed in percentage of increase in dielectric conductivity compared to conductivity by the control hydrogel.

Results and statistics

The measurements were performed in triplicates on each sample and for each time interval.
Statistical analysis was performed using one way ANOVA followed by PLSD Fisher test with a threshold of significance set 5% ($p<0.05$).

EFFICACY

Measuring dissemination and storage of serine and urea in the *stratum corneum*

ex vivo

OBJECTIVE

The purpose of this study was to measure the gradual dissemination and storage of the urea and serine molecules contained in PatcH$_2$O® at the level of the *stratum corneum* on human skin explants.
In this study the dissemination of free active ingredients was compared with those encapsulated in PatcH$_2$O®.

RESULTS & DISCUSSION

1. Storage of serine in the *stratum corneum*

Storage of serine in the *stratum corneum* is significantly increased in skin treated with PatcH$_2$O®, and this at all times (Figure 11).

2. Storage of urea in the *stratum corneum*

Storage of urea in the *stratum corneum* is significantly increased in skin treated with PatcH$_2$O®, and this at all times (Figure 12).

This study demonstrates that PatcH$_2$O® significantly increases the storage of urea and serine in the *stratum corneum* up to 48 hours after application.

In other words, this study shows that assets collected in the PatcH$_2$O® mesh are released gradually into the *stratum corneum* and that their stores in the upper layers of the epidermis are consequently increased.

The hydration can be maintained for up to 48 hours.

CONCLUSION

PatcH$_2$O® allows maintenance of active ingredients in the superficial layers of the skin, leading to long-lasting hydration.

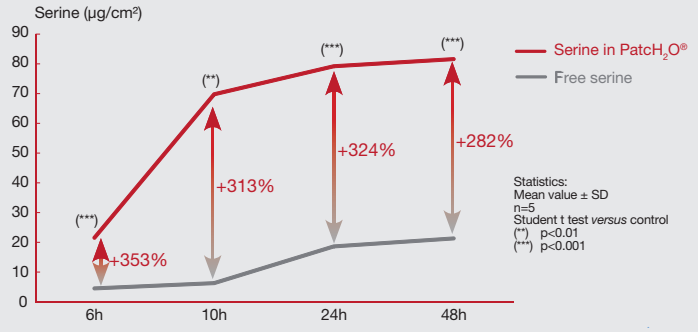


Figure 11 - Measurement of storage of serine in the *stratum corneum*.

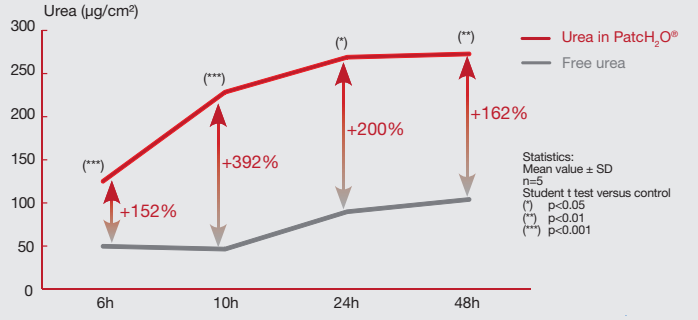


Figure 12 - Measurement of urea stored in the *stratum corneum*.

MATERIALS&METHODS

Material

Human skin explants were placed on Franz diffusion cells and equilibrated for 14 hours with phosphate buffered saline (PBS) containing antibiotics. 300 µl of the products to be tested were then added to the "donor" compartments of the cells, PBS solution was then added to the "receptor" compartments.



Franz cell

Treatment

Skin explants were incubated in the presence of serine and free urea (final concentration of 3% and 10%) and PatcH$_2$O® containing active ingredients at the same concentration.

300 µl of PatcH$_2$O® and moisturizing complex were applied to human skin biopsies. The treated skins were placed on Franz cells and incubated for 48 hours at 37°C.

Assay

At various times (6, 10, 24 and 48 hours), the skin biopsies were recovered, stripped (D-Squam®, CuDerm) and the quantities of serine and urea present in the *stratum corneum* were measured using HPLC.
All experiments were performed in quintuplicate (n = 5).

Results and statistics

The results expressed in mg of serine and urea/cm², are given in the form of the mean ± standard deviation (SD). Statistical significance was assessed using Student's t test with a threshold of significance set to 5% (p<0.05).

EFFICACY

Evaluation of PatchH_2O® moisturizing efficiency

This assessment took place in two separate clinical studies measuring the hydration of the upper layers of the epidermis: a study on the forearm after a single application and another half-face study after repeated applications during 21 days and then measuring the residual effect of hydration after stopping treatment.

Study on the forearm

in vivo

OBJECTIVE

The purpose of this study was to evaluate and compare the instant moisturizing effectiveness of PatchH_2O® *versus* a placebo after a single application on the forearm of healthy volunteers using the corneometry technique.

RESULTS & DISCUSSION

From 30 min to 48 hours after a single application, measures of hydration at all times for the formulation containing 3% PatchH_2O® were significantly higher compared to baseline. In addition, between 30 minutes and 24 hours, the moisturizing effect of PatchH_2O® was significantly better than that observed for the placebo ($p<0.05$). At 48 hours after single application on the forearm, the moisturizing effect of PatchH_2O® (33%) remained higher than that observed for the placebo (28%) (Figure 13).

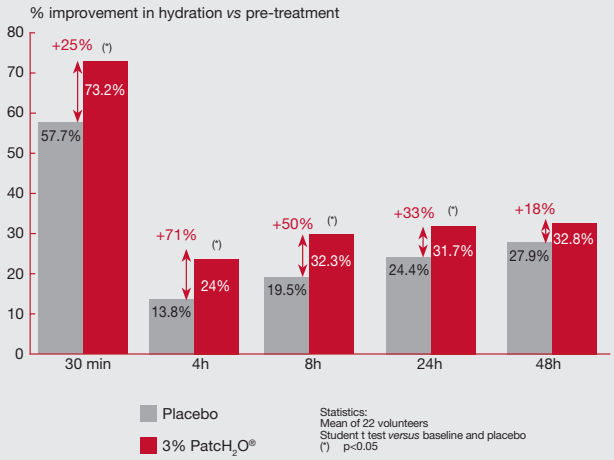


Figure 13 - *In vivo* study by corneometry of PatchH_2O® moisturizing contribution over 48 hours.

CONCLUSION

This study of 22 women shows that the formula containing 3% PatchH_2O® is an effective moisturizer of dry and very dry skin from 30 minutes to at least 48 hours after a single application to the forearm. This efficiency is significantly better than that of the placebo by at least 24 hours.

MATERIALS&METHODS

Study design

The clinical study was carried-out as a placebo-controlled double-blind randomized forearm study. The efficacy of the formulation containing PatchH_2O® at 3% was compared to the baseline (before treatment, D0) and to the other forearm treated with placebo. The study was conducted during a period of 48 hours with check points at at 30 min, 4, 8, 24 and 48 hours after a single application.

The used formulations, PatchH_2O® at 3% and Placebo (for detailed composition, cf Annex 2), differ only by the presence or not of PatchH_2O®.

This study was carried out from 6 to 22 March 2012 in BASF BCS clinical facilities of investigation based in USA.

Inclusion criteria

22 women aged between 18 and 65 years old, presenting a very dry skin (40 a.u. in corneometry or less) were recruited to participate in this study.

Application modality

3 days before the application of the test products volunteers followed pre-treatment by washing each forearm daily with Ivory Soap®.

The unique and standardized application of products (approximately 40 µl - (3 x 3 cm^2)) was performed on the forearm of volunteers by an experienced technician in a room with controlled temperature and relative humidity.

Evaluation methods

Hydration measurements were taken using a corneometer (Courage & Kazaka) in triplicate on each site before application (T0, baseline) and at 30 min, 4, 8, 24 and 48 hours after a single application.

The Corneometer uses the capacitance method to assess skin hydration. The capacitance method is based on the dielectric constant of water, measuring the change in capacitance according to the moisture content. An electric scatter field penetrates the skin during the measurement and the dielectricity is determined.

Additionally, Corneometer values of hydration below 40 a.u. (arbitrary units) correspond to very dry skin; between 40 and 55, dry skin; and higher than 55 normal skin[47].

Results and statistics

The results were expressed as mean values and percentages compared to baseline.

The statistical significance was assessed using Student's t test for paired samples with a significance threshold set at 5% ($p<0.05$).

Study on the face

in vivo

OBJECTIVE

The purpose of this study was to evaluate and compare the long lasting moisturizing effectiveness of PatcH$_2$O® *versus* a placebo after repeated applications on the faces of healthy volunteers using the corneometry technique.

After 21 days of treatment, measurements were also carried out over a period of time without applying products to assess the persistence of the moisturizing effect. In parallel, the volunteers were asked to complete a satisfaction questionnaire to self-assess the moisturizing qualities of the products tested and their effects on their skin.

RESULTS & DISCUSSION

Evaluation of moisturization effect

The results of this study showed that the placebo formulation presented no significant moisturizing effect compared to baseline (+14% and +12% *vs* baseline at times T23 and T26, respectively).

In contrast, 2 and 5 days after the last application, the product containing 3% PatcH$_2$O® showed a significant long-term moisturizing effect compared to baseline (+80% and +38% *vs* baseline at endpoint T23 and T26, respectively, $p<0.05$). PatcH$_2$O® was directionally better than the placebo (+66% and +26% respectively, $p<0.1$) (Figure 14).

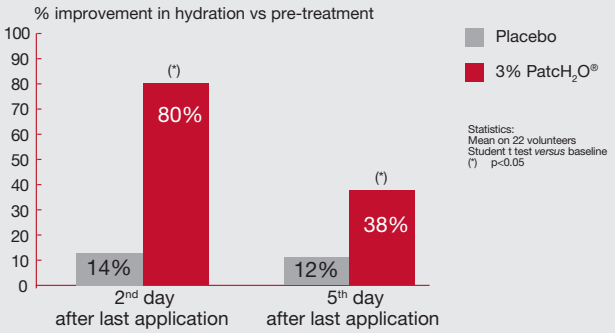


Figure 14 - *In vivo* study by corneometry of PatcH$_2$O® moisturizing lasting effect.

Self assessment

After 21 days of treatment, 72% of volunteers found that their skin "seems nourished" on the side treated with the product containing PatcH$_2$O® against 60% satisfaction for placebo (not significant, $p>0.05$).

After 21 days of treatment, the volunteers mostly acclaimed product containing PatcH$_2$O® and the number of volunteers satisfied with the effect of PatcH$_2$O® was superior to the placebo for the following items (Figure 15):

- skin hydrated
- softer skin,
- skin visibly more elastic,
- skin visibly nourished.

2 days after the last application, the volunteers felt that PatcH$_2$O® left their skin more hydrated than the side treated with placebo and 64% of them preferred the product containing PatcH$_2$O®.

5 days after the last application, the volunteers felt that PatcH$_2$O® left their skin more hydrated and more comfortable than the side treated with the placebo.

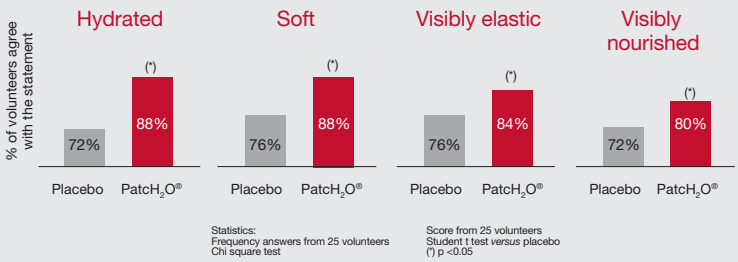


Figure 15 - Self-assessment by volunteers of moisturizing effect of PatcH$_2$O® after 21 days of treatment.

CONCLUSION

After 21 days of treatment, the formula containing 3% PatcH$_2$O® showed an intense moisturizing of the facial skin, the effect of which significantly persisted at least 5 days after the last application compared to T0 and placebo. In contrast, long-term moisturizing effect from the placebo after discontinuation of treatment, has not been demonstrated.

The effectiveness of PatcH$_2$O® has been confirmed by the evaluation of the volunteers themselves who judged, in the application of their products, that the skin treated with PatcH$_2$O® was more hydrated and nourished than with the placebo.

On most items, the volunteers were able to perceive a difference in favor of the product containing PatcH$_2$O®. We can also note that the beneficial effects of hydration and comfort were felt up to 5 days after discontinuation of treatment.

MATERIALS&METHODS

Study design

This study was carried out as a randomized, split-face, double blinded study in order to evaluate the PatchH_2O® formula compared to a placebo formula.

The used formulations, PatchH_2O® at 3% and Placebo (for detailed composition, cf Annex 2), differ only by the presence or not of PatchH_2O®. Treatment was stopped from T22 days inclusive, while measurements of hydration of the cheeks continued to T23 and T26 days to evaluate persistence.

Inclusion criteria

25 women aged between 18 and 65 years old, with very dry skin (40 a.u. of corneometer or less) were recruited to participate in this study, which took place from April 4 to June 20 BASF clinical facility located in US.

Application modality

Volunteers were given instructions and Ivory® soap for the 3 day washout phase.

Approximately 30µL of each formula was applied to the respective half of the face, twice daily, once in the morning and again in the evening for 20 days and only once the morning of the 21st day. From Day 22 until the final day of the study, Day 26, no product was applied to the face.

Evaluation methods

Hydration measurements were taken with a corneometer (Courage & Kazaka) in triplicate at each site on the cheeks (22 of 25 panelists).

Volunteers completed the satisfaction questionnaire on perceived benefits at T15 and T21 at home. For each item, the response options were "Strongly agree", "Agree", "Somewhat Agree," "Not really in Agreement", "Disagree", "Not all in Agreement"," No opinion".

Then at T23 and T26 volunteers rated their perception in terms of hydration and skin comfort at the test center by using a rating scale of 10 points (1: not at all hydrated/comfortable 10: extremely hydrated/comfortable).

Throughout the study period, the volunteers used a Neutrogena® soap in place of their usual product.

Results and statistics

The results were expressed as mean values and percentages *versus* baseline.

The statistical significance was assessed using Student's t test for paired samples with a significance level set at 5% ($p<0.05$).

The number of positive responses ("Strongly agree", "agree", "Somewhat agree") and negative ("Not really in Agreement", "Disagree", "Not all in Agreement" were grouped and a statistical analysis of the frequency of responses was performed (Chi square test, $p <0.05$).

A paired Student test was performed on the scores to compare results between PatchH_2O® and the placebo.



GENERAL CONCLUSION

PatcH$_2$O® is a breakthrough technology with anti-pollution effectiveness and immediate and long term moisturizing effects proven *in vivo*.

PatcH$_2$O™ is a unique association of natural polymers combined with a carefully selected and subtly dosed moisturizing complex, that bring the skin the appropriate moisture.

Through its unique combination of film-forming polymers, PatcH$_2$O® promotes the maintenance of the water molecules in the skin for 48 hours (models of Vitro-skin®) and demonstrate the capacity to offer prolonged hydration. *In vivo*, after just one application of PatcH$_2$O® at 3%, very dry skin recovers moisturization and remains significantly moisturized and reinforced for up to 48 hours compared to the baseline. After 3 weeks of 3% of PatcH$_2$O® use, moisturizing is maintained even up to 5 days after the last application.

Thanks to its film forming polymer, PatcH$_2$O® softly lays on the skin to form a barrier that limits penetration of pollutant in cells and thus their deleterious effects. Thus, PatcH$_2$O® can efficiently protects our skin cells from the cytotoxic effect and oxidative stress induced by UVA and some pollutant of industrialized city such as PM2.5 and BaP. Overall, PatcH$_2$O® also protects the skin from detrimental effect of pollutant deposition and adherence by increasing their removal with water. Its actions on preserving the skin beneficial microbiota against PM2.5 was also proven for an healthy balanced skin.

PatcH$_2$O®: the hydra-protect technology, a hydric barrier to fight pollution.

ANNEX 1

Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet

ANNEX 2

Clinical test formula

Trade name	INCI name	Placebo formulation %	$PatcH_2O^®$ formulation %
Water	Water	qsf 100	qsf 100
Disodium EDTA	Edeta® BD	0.05	0.05
Keltrol CG	Xanthan gum	0.20	0.20
Eumulgin® S2	Steareth-2	2.00	2.00
Eumulgin® S21	Steareth-21	2.50	2.50
Lanette® O	Cetearyl Alcohol	1.00	1.00
Cetiol® Sensoft	Propylheptyl Caprylate	15.00	15.00
$PatcH_2O^®$	Water, Glycerin, Trehalose, Urea, Serine, Pentylene Glycol, Glyceryl Polyacrylate, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	0.00	3.00
Sodium hydroxide	Sodium hydroxide, water (30% solution)	0.10	0.10
Germazide® PSB	Phenoxyethanol, Chlorphenesin, Benzoic Acid, Butylene Glycol, Sorbic Acid	1.25	1.25
Sepigel SMS 88 (Seppic)	Polyacrylate-X & Isohexadecane & Polysorbate 60	4.00	4.00



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